

View Review

Paper ID

772

Paper Title

Blockchain-Powered Trust Framework for Securing Online Examination Integrity and Authenticity Through Smart Validation

REVIEW QUESTIONS

1. Comments to Authors

□ The suggested system of trust based on a blockchain will fully cover online tests with an intends to establish biometric and behavioral verification combined with automation of smart contracts. The study successfully responds to the very crucial issue of integrities about exams, authenticating examinees, and keeping the scores credible through remote test settings.

□ The deep learning model that was developed with the system exhibits that it performs better than typical machine learning models such as Random Forest, Logistic Regression, and SVM.

Rather high quantitative evidence of the effectiveness of the model is presented on the basis of reported metrics, which are an accuracy of 92%, an F1-score of 0.92 and an ROC-AUC of 0.95.

□ The analysis of statistical significance is one of the strong points of this piece since it ensures that the gains presented with regard to the proposed model are not a fortuitous occurrence.

The very small p-value of ultra-confirms the model in its strength, validly improving on the weakest baseline.

□ It is a remarkable limitation to resort to the synthetic dataset, as is admitted by the authors, and cannot precisely reflect the opportunities and unpredictability that real-life online examinations hold. Additional research ought to be done to confirm the presented framework to real-world settings to ensure its reliability and generalizability in a variety of settings.

□ The figures are not represented visually very well. According to Figure 2, the ROC curve has contradicting values of the AUC compared to the text, and the confusion matrices presented in

Figure 3 are small to interpret. The following are areas to be done to make the results more readable and employable in their credibility.

2. Recommendation: Your overall rating

• Minor Revision
